ECON100B - SECTION 2

MATH REVIEW and GDP BASICS

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Credit to Kevin Yin, Anubhava Singla, & other previous 100B GSIs for parts of this lesson

AGENDA

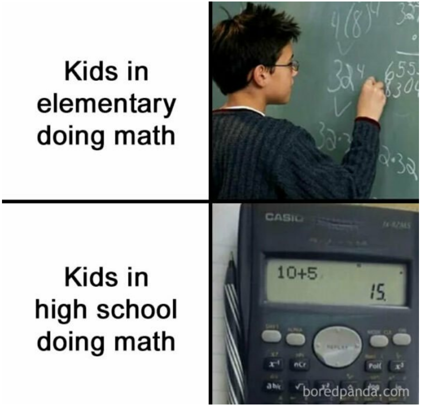
- Math Review
- GDP Review

Question of the Day

1. If I am a country that produces $20M in steel which is used to produce $100M in car bodies, which is used to produce $200M worth of cars, what is my GDP?

2. If you have extra time, what is your least favorite food?

Math Review 💃


Kids in elementary doing math
Kids in high school doing math
CASIO
10+5
15
boredpanda.com

MATH REVIEW: PERCENTAGES

Percentage point vs percentage changes

- Percentage point change measures the absolute difference between two percentages
- A jump from 20% to 30% is a 10-percentage point change

- Percentage change measures the relative change between two percentages
- A jump from 20% to 30% is a 50-percent change

MATH REVIEW: PERCENTAGE CHANGE

There are two ways to compute percentage changes:

- Calculate it directly:

$$\%\Delta x_t = 100 \times \frac{x_{t+1} - x_t}{x_t}$$

- Approximate it using logs:

$$\%\Delta x_t \approx 100 \times (\ln(x_{t+1}) - \ln(x_t))$$

MATH REVIEW: PERCENTAGE CHANGE CONT.

Why does the log approximation work?

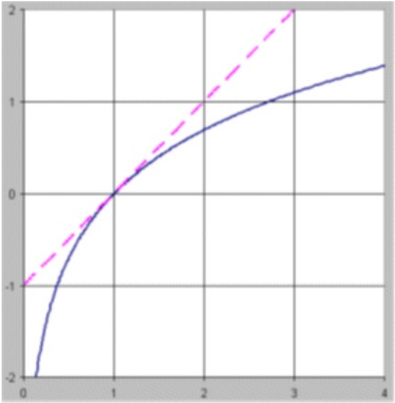
- Remember from calculus - Taylor approximation
- Lets you approximate a function around a point x_0

$$
\begin{aligned}
f(x) &\approx f(x_0) + f'(x_0)(x - x_0) \\
\Longrightarrow \ln(x) &\approx \ln(x_0) + \frac{1}{x_0}(x - x_0) \\
\Longrightarrow \ln(x_{t+1}) &\approx \ln(x_t) + \frac{1}{x_t}(x_{t+1} - x_t) \\
\Longrightarrow \ln(x_{t+1}) - \ln(x_t) &\approx \frac{x_{t+1} - x_t}{x_t}
\end{aligned}
$$

MATH REVIEW: PERCENTAGE CHANGE CONT.

- That math was too confusing …
- Here's another way to look at it:

- Every percentage change is a fraction – most small percentage changes will have a fraction close to 0: for example: (4.00-3.75)/3.75 = .07
- We can model all percentage changes with a straight-line $y = x – 1$
- The function $y = \ln(x)$ very closely follows the straight-line at $x = 1$, so for all small percentage changes around 0, the natural log is a good approximation!



PRACTICE

Suppose Japan's GDP was 5.1 trillion USD in 2008 and grew to 6.2 trillion in 2011. What was the percentage change from 2008 to 2011?

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Suppose Japan's GDP was 5.1 trillion USD in 2008 and grew to 6.2 trillion in 2011. What was the percentage change from 2008 to 2011?

- Calculate it directly:

$$\%\Delta x_t = 100 \times \frac{6.2 - 5.1}{5.1}$$

- Approximate it using logs:

$$\%\Delta x_t \approx 100 \times (\ln(6.2) - \ln(5.1))$$

PRACTICE

Suppose Japan's GDP was 5.1 trillion USD in 2008 and grew to 6.2 trillion in 2011. What was the percentage change from 2008 to 2011?

- Calculate it directly:

$$\%\Delta x_t = 21.57\%$$

- Approximate it using logs:

$$\%\Delta x_t \approx 19.53\%$$

MATH REVIEW: LOGARITHM RULES

Logs are useful for other reasons too

- They turn multiplication into addition

$$\ln(x\,y) = \ln(x) + \ln(y)$$

$$\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$$

- They turn exponents into multiplication

$$\ln(x^a) = a\ln(x)$$

So why are we talking about logs?

- A lot of the models that we use in (macro)economics are multiplicative
- Models have exponents, which we don't want to work with (because that math is way too hard!)

Consider the following:

$$y = Ax^{\beta}$$

How do you solve or interpret what's going on here? Use logs:

$$\ln(y) = \ln(A) + \beta\ln(x)$$
$$\beta = \frac{\partial ln(y)}{\partial ln(x)}$$

This looks like a familiar linear regression, akin to the form Y = mx + b! Which we can interpret as an elasticity: a 1% change in x changes y by beta percent!

MATH REVIEW: DERIVATIVES

Basic rules:

- $\frac{d}{dx} cx =$
- $\frac{d}{dx} x^n =$
- $\frac{d}{dx} \ln(x) =$
- $\frac{d}{dx} e^x =$

Addition, product and quotient rules:

- $\frac{d}{dx} f(x) + g(x) =$
- $\frac{d}{dx} f(x)g(x) =$
- $\frac{d}{dx} \frac{f(x)}{g(x)} =$

MATH REVIEW: DERIVATIVES

Basic rules:

- $\frac{d}{dx}cx = c$
- $\frac{d}{dx}x^n = nx^{n-1}$
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- $\frac{d}{dx}f(x) + g(x) = f'(x) + g'(x)$
- $\frac{d}{dx}f(x)g(x) = f'(x)g(x) + g'(x)f(x)$
- $\frac{d}{dx}\frac{f(x)}{g(x)} = \frac{f'(x)g(x) - g'(x)f(x)}{g(x)^2}$

Why do we have to talk about derivatives in this class ...

- Optimization of functions – maximization 🤑 !
- Want to see how we can maximize the output of different functions/models when we change parameters (comparative statics)
- Partial derivatives help us understand the influence of different variables for outcomes that we're interested in!

PRACTICE

In groups, solve the following:

1. An economy has experienced a change in unemployment rate from 4.0 percent to 5.2 percent in the past year. What are the percentage point and percentage changes in the unemployment rate in this economy?
2. The country Oski-land has seen net exports rise from $47M to $52M in the past year. Calculate the percentage change in net exports using the direct and log calculation techniques
3. Simplify the following: $ln(2xy^3)$
4. Take the following derivative: $\frac{\partial \mathrm{Y}}{\partial \mathrm{K}}$ for equation $Y = K^{\alpha}L^{1-\alpha}$
5. Take the following derivative: $\frac{\partial}{\partial \mathrm{x}} ln(x^2y)$

MATH REVIEW WRAP UP:

- For some of you may be excited to use math in this class! For others of you this might be terrible news ☹

- Our goal is to understand the intuition of our models – math helps us do that – you don't need to understand all the intuition behind the math, just how to use it!
- We'll have plenty of time to talk about things that are not math related in class as well!

- If you have any questions about problem solving approaches during this class, please reach out to me by email of during OH, and/or collaborate with peers to build intuition!

GROSS DOMESTIC PRODUCT - MEASUREMENT

There are two ways we actually measure GDP in reality:

1. Expenditure approach - add up all the spending in on **domestically produced final goods** in the economy
2. Income approach - add up all the earnings on **domestically located** factors of production (capital and labor, e.g. a wage and the surplus of a company)

EXPENDITURE BREAKDOWN

$$Y = C + I + G + NX$$

To understand this equation, ask yourself, what kinds of things can we do with the resources we create?

- People could use it up, ie. consume it (*C*)
- The government could use it up, ie. consume it (*G*)
- We could save it for later or use it to make more stuff, ie. invest it (*I*)
- We could sell it abroad, ie. export it (*NX*)

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CONSUMPTION

Consumption (*C*) is the resources used up by households

- Non-durable goods
- Durable goods
- Services

WHAT COUNTS AS INVESTMENT EXACTLY?

- Resources spent on more capital (being “put to work")
- **In a closed economy**, any money people save gets invested
- The money you save in your bank account gets lent by the bank to companies/firms
- Investment (*I*) does not have to be in financial securities; leftover inventory is investment too because it’s resources that are not being used this year

GOVERNMENT SPENDING

Government spending *G* is really government “consumption”, in that it’s the resources the government, not households, have used up.

What counts as *G*?

What’s part of the government’s budget but doesn’t count as *G*?

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- **Hiring bureacrats to run different departments**
- **Paying for military equipment and soldiers**
- **Materials and labor for building infrastructure**

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What's part of the government's budget but doesn't count as G?

- **Transfer payments**

PRACTICE

Q: What are transfer payments? Why are they called that?

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Transfer payments are any redistribution of income from one group to another, through the government. They are called transfers because they just move wealth around without creating any.

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Q: Why do they not count toward government spending?

Transfers only move money from one group that would've spent the money, to another group that will then spend the money, which either way is counted in final consumption.

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Net exports are thus all the money foreign countries spent on our goods and services (gross exports), net of the money we spent on theirs (gross imports).

THINKING CRITICALLY ABOUT GDP

PEACE

The Problem with GDP

Published On: May 6, 2022

In this extract from his debut book, ‘Peace in the Age of Chaos’, IEP Founder and Executive Chairman Steve Killelea unpacks the problem with GDP and why we need better measures of societal wellbeing and progress.

As many have pointed out, if businesses used GDP-style accounting, they would aim to maximise gross revenue at the expense of profitability, efficiency, sustainability and flexibility.

GDP makes no additions to take into account the health of the population, the quality of education, the strength of social relationships, the intelligence of public debate or the integrity of public institutions, nor subtractions to account for social strife, inequality or environmental degradation. If a house burnt down, then a new house would need to be built. This would increase purchases of building materials and require labour to rebuild it, thereby increasing GDP. Viewed through this distorted lens, it would be considered a better outcome than if the house had not burnt down.

However, there is a net loss in the quality of life. In the same way, many of the elements of violence, such as expenditure on the military, jails, policing, the judiciary and security will show up in GDP as a positive indicator, a way in which the economy has ‘grown’. Because GDP is a measure of the money being exchanged, and not what it is being exchanged for, it creates a deeply distorted, materialistic picture of what a society is. When there is a heavy focus on GDP there is also a tendency to reinforce the status quo, the existing transactional system. This is especially obvious with the environment, where heavily polluting activities are seen as a benefit to GDP because there is already an established set of transactions related to them, such as income from electricity generated by coal-fired power stations.

Think-pair share: Thinking Critically About GDP

- What does the author think is the main issue with existing measures of welfare?
- Do you agree? Why or why not?
- If you had to respond to Killelea in defense of GDP as a measure, what would you say?